

Smart Air Quality MONITORING SYSTEM

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Air pollution is a critical concern in modern living environments, particularly in enclosed spaces such as homes, offices, factories, and laboratories, where prolonged exposure to harmful gases presents serious health risks. Continuous air quality monitoring is therefore essential to maintain safe and healthy indoor conditions. Here we detail the design and implementation of a compact, low-cost, smart air-quality monitoring system based on the Arduino Nano platform.

The system employs an MQ135 gas sensor to detect harmful gases in ambient air continuously. Based on predefined threshold levels, air quality is categorised as good, poor, very poor, or toxic and displayed on a 16×2 LCD with an I²C interface. An audible alert is generated by a buzzer connected to a digital output pin, which activates when the air quality exceeds safe limits.

Although a standardised air quality index (AQI) is not computed, the system effectively indicates pollution levels and provides real-time alerts. Fig. 1 shows the system prototype, and the Bill of Materials table lists the required components.

Circuit and working

The circuit diagram of the Arduino Nano-based air quality monitoring system is shown in Fig. 2. The system is built around an Arduino Nano (MOD1), an MQ135 gas sensor module (S1), a 16×2 LCD with an I²C interface (MOD2), a buzzer, and a small number of additional components.

The system can be powered either through the 5V Vin pin or

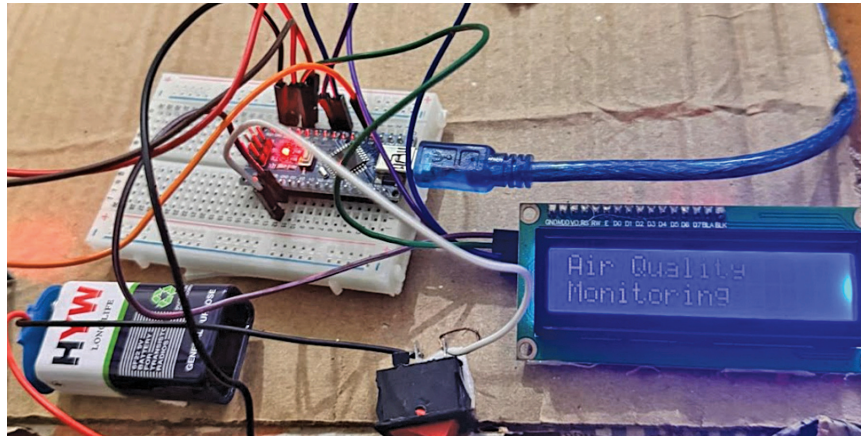


Fig. 1: Author's prototype

via a USB cable. Both the MQ135 sensor and the LCD are powered by the Arduino Nano's 5V regulated output. The LCD connects to the Arduino Nano's I²C pins, while the gas sensor connects to the analogue pin A1. The buzzer is connected to digital pin D2.

The Arduino Nano serves as the system's central processing unit. It reads the sensor output, processes the data, and controls the display and alert mechanisms. The MQ135 gas sensor module detects harmful gases such as ammonia, nitrogen oxides, smoke, benzene, and carbon monoxide. The sensor operates on the principle that the resistance of its sensing element changes in the presence of different gases, producing an analogue output voltage proportional to the gas concentration.

The analogue output (AO) of the MQ135 sensor is connected to the Arduino Nano's A1 analogue input pin, while the Arduino's 5V supply powers the sensor module. The continuously varying analogue signal from the sensor is converted

BILL OF MATERIALS

Compnents	Quantity
Arduino Nano (MOD1)	1
16x2 LCD I2C (MOD2)	1
MQ135 (S1)	1
Buzzer (B1)	1
USB cable	1
Jumper wires	40
Breadboard	1
9V battery (optional)	1

into a digital value by the Arduino's internal analogue-to-digital converter (ADC).

A 16×2 LCD module with an I²C interface is used to display air quality readings. The I²C interface minimises wiring complexity by using only two communication lines. The SDA and SCL pins of the LCD are connected to the Arduino Nano's A4 and A5 pins, respectively. The LCD displays real-time air quality data in a clear and readable format, enabling easy monitoring.

A buzzer connected to the D2 pin provides an audible alert. The Arduino firmware continuously compares the sensor reading against